FOR A FLAT MORPHISM GORENSTEINNESS AT A POINT IS EQUIVALENT TO THE SHRIEK BEING INVERTIBLE NEARBY

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file duality.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with an input of the shared preamble and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-duality/preamble.tex is excerpted verbatim under the recipe fixed by LANE_BRIEF 22.1, as three spans, lines 8-13 (verbatim package and the reference/slogan/history environments), lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 0C08 (sources/stacks-duality/pinned-original-source.tex line 6722).

Allowed standard prerequisites (editorial).

- The shriek of the structure sheaf is a bounded complex with coherent cohomology whose fibre restriction is dualizing. Same chapter, Lemma labelled lemma-relative-dualizing-fibres, cited at proof lines 6723-6727. Makes the derived restriction to the fibre a dualizing complex.
- The criterion for Gorensteinness through the dualizing complex of the fibre. Same chapter, Lemma labelled lemma-gorenstein, cited at proof lines 6749-6751. Gives the equivalence of Gorensteinness and invertibility on the fibre.
- Lifting invertibility from the point to a neighbourhood. More on Algebra, Lemma labelled more-algebra-lemma-lift-bounded-pseudo-coherent-to-perfect, cited at proof lines 6753-6757 and 6759-6763. Moves between the dimension one condition at the point and invertibility nearby.

Source reference. Original reference 0F42: sources/stacks-duality/pinned-original-source.tex, source lines 3953–3953. Same-chapter pointer into the author environment situation at source line 3953; invoked at proof line None of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0BZZ: sources/stacks-duality/pinned-original-source.tex, source lines 5130–5130. Same-chapter pointer into the author environment lemma at source line 5130; invoked at proof line 6737 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Date: stacks-project:duality.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 0BFQ: sources/stacks-duality/pinned-original-source.tex, source lines 6400–6400. Same-chapter pointer into the author environment lemma at source line 6400; invoked at proof line 6755 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0BYP: sources/stacks-chow/tags.txt, source lines 15022–15022. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 6758 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *In Situation 0F42 let $f : X \to Y$ be a morphism of FTS_S. Let $x \in X$. If f is flat, then the following are equivalent*

(1) *f is Gorenstein at x,*
(2) *$f^!\mathcal{O}_Y$ is isomorphic to an invertible object in a neighbourhood of x.*

In particular, the set of points where f is Gorenstein is open in X.

Proof. Set $\omega^\bullet = f^!\mathcal{O}_Y$. By Lemma 0BZZ this is a bounded complex with coherent cohomology sheaves whose derived restriction $Lh^*\omega^\bullet$ to the fibre X_y is a dualizing complex on X_y. Denote $i : x \to X_y$ the inclusion of a point. Then the following are equivalent

(1) f is Gorenstein at x,
(2) $\mathcal{O}_{X_y,x}$ is Gorenstein,
(3) $Lh^*\omega^\bullet$ is invertible in a neighbourhood of x,
(4) $Li^*Lh^*\omega^\bullet$ has exactly one nonzero cohomology of dimension 1 over $\kappa(x)$,
(5) $L(h \circ i)^*\omega^\bullet$ has exactly one nonzero cohomology of dimension 1 over $\kappa(x)$,
(6) $\omega^\bullet$ is invertible in a neighbourhood of x.

The equivalence of (1) and (2) is by definition (as f is flat). The equivalence of (2) and (3) follows from Lemma 0BFQ. The equivalence of (3) and (4) follows from More on Algebra, Lemma 0BYP. The equivalence of (4) and (5) holds because $Li^*Lh^* = L(h \circ i)^*$. The equivalence of (5) and (6) holds by More on Algebra, Lemma 0BYP. Thus the lemma is clear. □